Using the Analytic Hierarchy Process for ABC Analysis

Fariborz Y. Partovi and Jonathan Burton
Department of Management, Drexel University, Philadelphia, USA

Received August 1992
Accepted December 1992

Introduction

Thousands of items may be held in inventory by a typical organization, but only a small percentage of them deserve management's closest attention and tightest control. ABC analysis allows organizations to separate inventory items into three classes: A, very important; B, moderately important; and C, least important. Only class A items deserve management's utmost attention. The classification of an item into the A, B, or C category has generally been based on a single criterion: dollar value per unit multiplied by annual usage rate, commonly known as dollar usage[1]. For many items, however, there may be other criteria which represent important considerations for management. The certainty of supply, the rate of obsolescence, the criticality of the item, and the impact of stock-out of the item are all examples of such considerations. Some of these may even weigh more heavily than dollar usage in the procurement planning and control of the item.

This article proposes a multicriteria model for classifying inventory items. The proposed model is based on Saaty's Analytic Hierarchy Process (AHP)[2], which contributes to the assignment of inventory items to A, B, and C categories. The article first provides a summary of prior research on the multi-criteria ABC classification. Second, it briefly describes AHP and, third, discusses the proposed AHP model. Next, data from a pharmaceutical company are used to verify the practicality and effectiveness of the model. Finally, the article concludes with a summary and implications for further study.

Background

Although much has been written criticizing the use of the traditional dollar usage classification method (see, for example,[3,4]), only recently have researchers proposed prescriptive methodologies to replace it. For example, Flores and Whybark[5,6] have proposed the bi-criteria matrix approach. Essentially, their approach is to use standard ABC analysis on each of two criteria and then combine the two single-criterion ABC groupings through the use of a joint-criteria matrix. Though this approach is a step forward in multicriteria ABC analysis, it does have some limitations. First, there is no obvious way to extend the procedure to more than two criteria. Second, the weights of different criteria are assumed to be equal.

Another pair of researchers, Ernst and Cohen[7,8], have proposed a new methodology based on statistical clustering, which utilizes a full range of operationally significant attributes. The main advantage of this approach is that it can accommodate large numbers of combinations of attributes which are

International Journal of Operations & Production Management, Vol. 13 No. 9, 1993, pp. 29-44. © MCB University Press, 0144-3577

Downloaded from http://www.emerald.com/ijopm/article-pdf/13/9/29/975952/01443579310043619.pdf by BI Norwegian Business School user on 26 February 2026
© Emerald Backfiles 2007

significant for both strategic and operational purposes. However, this requires substantial data, the use of factor analysis, and clustering procedures which may make it impractical in typical stockroom environments, where the decision makers might lack familiarity with the statistical procedures necessary to implement it properly. Furthermore, the clusters themselves must be re-evaluated in order to classify new stock items, so there is a chance that previously classified stock may end up being classified differently every time new stock items are added; this may disturb the inventory control procedures. In short, their model may be too sophisticated for the average manager.

This article introduces a computer model for ranking the importance of inventory items. The proposed model, which is based on Saaty's AHP, ranks inventory items based on several criteria. Furthermore, it does not give equal weights to these criteria. The proposed model assumes that inconsistencies may occur in qualitative judgements. By asking for pair-wise comparisons, inconsistencies are handled formally.

Analytic Hierarchy Process

AHP is a decision-support procedure for dealing with complex, unstructured, and multiple-attribute decisions. Since its initial development, AHP has been applied in a wide variety of decision areas, including those related to economic planning, energy policy, health, conflict resolution, project selection and budget allocation. (For an interesting review, see Zahedi[9].) Although many applications of AHP are in the area of socio-economic planning, there have been relatively few applications in decisions related to manufacturing/production systems[10-13]. A review, along with suggestions for new applications, is given by Partovi *et al.*[14].

There are three basic steps in using AHP:

(1) The description of a complex decision problem as a hierarchy.

(2) The use of pair-wise comparisons to estimate the relative weight (importance) of the various elements on each level of the hierarchy.

(3) The integration of these weights to develop an overall evaluation of decision alternatives.

In a typical hierarchy, the top level reflects the overall objective of the decision maker. The elements affecting the decision are called criteria, and they are represented at the intermediate levels. Criteria can be further divided into sub-criteria for additional refinement. The criteria can be subjective or objective, depending on the means used in evaluating the contribution of the elements below them in the hierarchy. Furthermore, criteria are mutually exclusive and do not depend on the elements below them in the hierarchy. The lowest level comprises the decision options or alternatives.

The number of criteria or alternatives should be reasonably small to allow consistent pair-wise comparisons. Saaty[2] suggests a maximum number of seven. In cases where the number of criteria or alternatives is more than seven, two possibilities exist. One can group the criteria with respect to a common property and add another level to incorporate the groupings[15,16]. In the

Downloaded from http://www.emerald.com/ijopm/article-pdf/13/9/29/975952/01443579310043619.pdf by BI Norwegian Business School user on 26 February 2026
© Emerald Backfiles 2007

case of the alternatives, the ratings mode of AHP may be used. In the ratings mode of AHP, in addition to the three general levels of: the objective; the criteria (sometimes subdivided into additional levels); and the alternatives, we have a level above the alternatives consisting of ratings which are refinements of the criteria or subcriteria governing the alternatives[16,17].

Once the hierarchy has been constructed, the decision maker begins the details of pair-wise comparisons to estimate the relative importance of the various elements on each level. The next step is the integration of these elements, weights for an overall prioritization of decision alternatives. These steps are by now relatively well understood using the established AHP procedures and supporting AHP software such as "Expert Choice"[18]. The reader interested in the details of these procedures is referred to Saaty[2].

An important benefit of AHP is that the process takes into account inconsistency in preferences. Inconsistencies exist because redundant information relating to priorities is present in each preference matrix. Methods are available for measuring the inconsistency of a pair-wise comparison matrix. If the inconsistency exceeds 0.10, some revisions of judgement may be required. More detail on preference matrices and consistency can be found in Saaty[2].

AHP can be used to make relative measurements through paired comparisons of criteria and of alternatives, or to make ratings measurements (scoring) of the alternatives with respect to the criteria[16,17]. The ratings mode includes pair-wise comparison of the criteria with respect to the goal. Then rating levels, such as very high, high-medium, low, very low, are specified for each criterion. Pair-wise comparisons among the rating levels of each criterion are then conducted to yield a set of priorities (weights) for these levels. For each criterion, the rating level priorities are divided by the maximum rating weight of that criterion to yield scaled weights[16]. Within each criterion, each alternative is assigned a rating level and the associated scaled weight. The final score for an alternative is the sum of the product of the criterion weight times the scaled weight with respect to that criterion, where the sum is taken across all criteria.

The ratings mode has been used when the number of alternatives is large (greater than seven) and decisions are standardized[16,17]. Furthermore, in using the ratings mode, no matter how many new alternatives are introduced, or old ones deleted, the ranks of the alternatives cannot be reversed[19]. Thus the question of rank reversal and its legitimacy, which is one of the more heavily discussed aspects of AHP, is not relevant in this mode. The only requirement for the ratings mode is having expert knowledge to be able to compare rating levels with respect to a certain criterion[20].

ABC Classification Methodology

This section proposes a procedure for prioritizing stock-keeping units (SKUs) for management attention based on AHP. The procedure rates inventory items, with respect to both qualitative and quantitative criteria. The final score obtained from AHP will be used to determine the cut-offs for each of the classes, A, B, and C. The AHP-based hierarchy for evaluating inventory items is illustrated in Figure 1.

Downloaded from http://www.emerald.com/ijopm/article-pdf/13/9/29/975952/01443579310043619.pdf by BI Norwegian Business School user on 26 February 2026
© Emerald Backfiles 2007

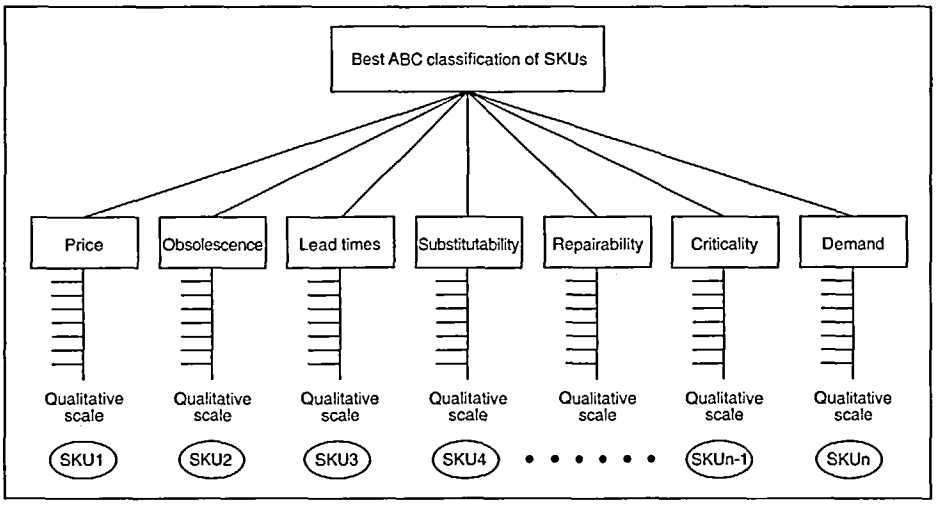

Figure 1.
Decision Hierarchy for ABC Classification of SKUs

The objective of the procedure is shown at the highest level of the hierarchy. At the second level, various criteria for ABC classification are presented. Flores and Whybark[5,6] have suggested price, obsolescence, lead times, substitutability, repairability, criticality, and demand. The important feature of which one needs to be aware is the independence of these criteria. If the decision maker feels that some criteria are not independent, then those criteria should be aggregated into a single criterion. Furthermore, criteria need to have both discriminating power and relevance. Criteria which have limited discriminating power will tend to cluster all items into one class. However, strong discriminators which are not relevant can cause improper clusters to be formed. Indeed, identifying the appropriate criteria probably has the greatest impact on classification of SKUs.

Once all criteria are defined, the evaluation process continues using the established procedures of AHP. At the first level of the hierarchy, decision makers are asked to specify the relative importance of each of the selected criteria in achieving the best SKU classification. For example, the questions could be posed as follows: "Assuming that our objective is to classify our SKUs into three groups, how important is obsolescence in comparison with lead times in this classification?" Decision makers can answer these questions with relative ease using descriptive preferences (e.g. equally important, moderately more important, etc.) discussed earlier. The outcome of this evaluation is a preference matrix presenting these judgements as numerical values. Once the preference matrix is determined, the standard AHP calculations are employed to determine the local priorities or weights for criteria.

The alternatives considered in this hierarchy are all the SKUs, as shown at the bottom of Figure 1. It is not practical to use the relative mode of AHP to compare each item with all other items with respect to each of the established criteria. Furthermore, AHP imposes a restriction on the maximum number of elements (seven) which can be pair-wise compared. Thus, in order to rate

Downloaded from http://www.emerald.com/ijopm/article-pdf/13/9/29/975952/01443579310043619.pdf by BI Norwegian Business School user on 26 February 2026
© Emerald Backfiles 2007

the importance of each SKU in relation to each of the established criteria, the ratings mode will be used (as represented by Figure 1, where the different rating levels for each criterion are introduced).

In the ratings mode, the comparisons of rating levels corresponding to each criterion will lead to weights or priorities for each of these rating levels. The procedure is as before, and the decision maker can pose questions such as: "In evaluating SKUs, how much more important is a high rating than a moderate rating for obsolescence?" These comparisons lead to weights for each of the three rating levels associated with each criterion. The weights are then scaled by dividing each weight by the largest weight within each criterion[16]. These scaled weights and their associated criteria then can be transferred to a spreadsheet so that a rating for each criterion can be selected for each SKU. The weights for the selected ratings are added for a total weight for each SKU. At this stage, the total weights thus calculated will be used for final classification of the SKUs into A, B, and C categories.

A Case Analysis

As a first step in testing this model, the authors proposed the model to a large pharmaceutical company, located in suburban Philadelphia, for classifying their maintenance department stockroom inventory. Company management expressed enthusiasm for the idea and authorized an initial study to compare their current dollar usage classification and the proposed AHP classification. The study was conducted as follows.

A sample of 96 items was taken, and is given in the Appendix. Each spare part observation contained four types of information: unit cost ($/unit), procurement cost ($/order), demand range (units/year) and lead time (days) (*Note*: for the sake of brevity and confidentiality, product codes have been substituted for actual item names.) The ABC dollar usage classification currently labels any item with a dollar usage cost of over $3,100 as an "A" item; any item with a dollar usage cost of between $400 and $3,099 as a "B" item; and all other items fall under the "C" category.

Management determined after careful consideration that four criteria were most important: criticality, lead time, unit price of part, and demand, as shown in the second level of Figure 2. Criticality is the word used to describe the consequence of not having an item on hand when it is required. Lead time is the elapsed time between placing an order with a vendor and when that order arrives. SKU price was included as a criterion for classification, because it directly affects the holding cost of the item and also increases idle capital. The final criterion for judging the SKUs is annual demand (usage). It is important to monitor and control high-demand items so that shortages can be reduced.

In order to develop the priorities of the criteria, management was asked to compare each pair of criteria. Expert Choice[18] software was then used to evaluate these comparisons. The outcome of this evaluation process was 0.500, 0.100, 0.200, and 0.200 for criticality, lead time, unit price of product, and demand, respectively. These weights are shown in Figure 2. The inconsistency ratio (CI) was 0.00.

Downloaded from http://www.emerald.com/ijopm/article-pdf/13/9/29/975952/01443579310043619.pdf by BI Norwegian Business School user on 26 February 2026
© Emerald Backfiles 2007

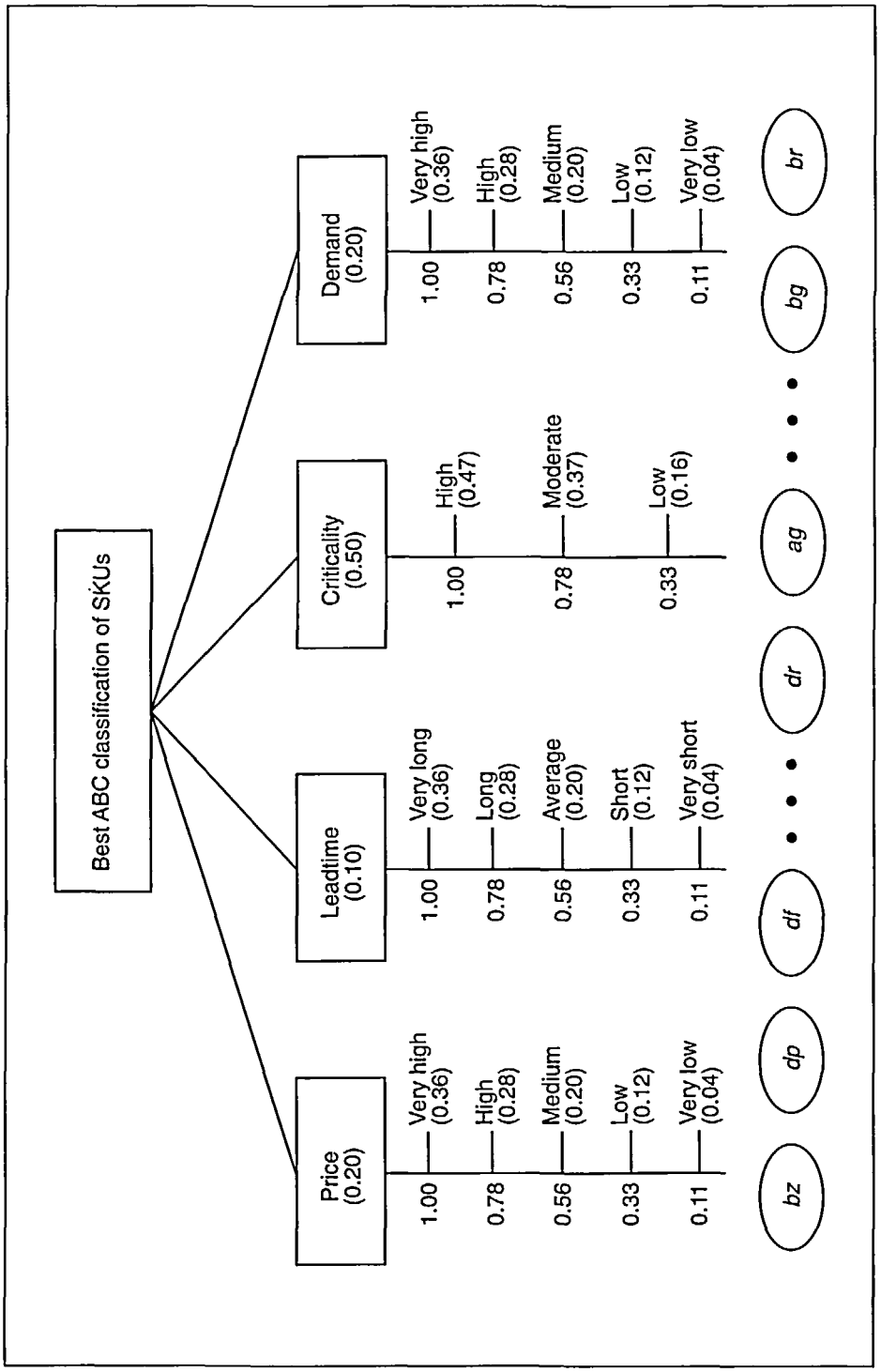

Figure 2.
Decision Hierarchy for ABC Classification of SKUs

Downloaded from http://www.emerald.com/ijopm/article-pdf/13/9/29/975952/01443579310043619.pdf by BI Norwegian Business School user on 26 February 2026
© Emerald Backfiles 2007

The level below that of the criteria, in Figure 2, consists of intensities or grades, which are refinements of the criteria. For criticality criteria, three rating levels were used: high, moderate, and low. If the absence of the SKU led to complete shutdown, then it was classified as highly critical. If the absence of the SKU led to partial shutdown, then it was classified as moderately critical.

Finally, if the absence of the SKU led to no real problem, then it was classified as being of low criticality. The weights of these ratings were obtained by management's pair-wise comparisons. The local weights of these ratings are 0.47, 0.37, 0.16, (C1 = 0.005), as shown in Figure 2. The corresponding scaled values are 1.00, 0.78, and 0.33, which were used for evaluating alternative SKUs in the second column of Table I. For price there were five categories, defined as very low, low, medium, high, and very high. The price range was 0-$4.49 for very low, $4.5-$19.99 for low, $20-$104.99 for medium, $105-$599.99 for high, and $600-up for very high. The local weights of these ratings and the corresponding scale values were calculated as before. The other two criteria, lead time and demand, were similarly analysed. The local weights and the corresponding scaled value of ratings for each criterion are shown in Figure 2.

One of the intensity levels associated with each criterion was used to evaluate the SKUs. For example, the evaluations of the part code *bz* with respect to price, lead time, criticality, and demand criteria are very low, very short, partial, and very high, respectively. The final scores for the SKU values are presented in the right column of Table I, and are found by summing the product of the criterion weights times the scaled rating level priorities for SKUs across all criteria. For example, the total value of the part code *bz* is obtained as follows:

$$(0.20)(0.11) + (0.10)(0.11) + (0.50)(0.33) + (0.20)(1.00) = 0.40$$

Since the object of this study was to demonstrate to management that AHP classification would be superior to the dollar usage classification currently in place, the study employed the same fixed percentages as the current dollar usage model to determine the number of SKUs in each ABC class. A items constituted approximately the top 15 per cent, B items approximately the next 35 per cent, and C items approximately the last 50 per cent of SKUs. In practice, however, the discontinuity of total AHP values provides a logical basis for separating Class A, Class B and Class C items.

Simulation

In order to show management the effects of using AHP classification over time, a simulation model was developed to compare the proposed AHP model with the traditional dollar usage classification. The demand for SKUs occurs when machines in the plant experience failure. A demand rate following a Poisson distribution with mean equal to the mid-range of the expected demand for each item was assumed. The time between demands for an item follows an exponential distribution with mean equal to the inverse of average demand. The lead time for each item is represented by a normal distribution with the mean equal to the mid-point of the upper and lower values of the expected lead time for that item, and standard deviation estimated by the demand range for each item divided

Downloaded from http://www.emerald.com/ijopm/article-pdf/13/9/29/975952/01443579310043619.pdf by BI Norwegian Business School user on 26 February 2026
© Emerald Backfiles 2007

Part code	Unit cost (0.20)	Lead time (0.10)	Downtime (0.50)	Demand (0.20)	Final score
bz	0.11	0.11	0.33	1.00	0.40
dp	0.33	0.33	0.78	0.78	0.64
df	1.00	1.00	1.00	0.11	0.82
cs	0.11	0.11	0.33	0.78	0.36
ax	0.56	0.33	0.33	0.56	0.42
bb	1.00	1.00	1.00	0.33	0.87
bv	0.11	0.11	0.33	0.78	0.36
bj	0.11	0.11	0.78	1.00	0.62
at	0.56	0.56	1.00	0.78	0.82
ab	0.33	0.56	0.78	1.00	0.71
du	0.78	0.56	1.00	0.33	0.78
dr	0.78	0.56	0.33	0.56	0.49
ag	0.11	0.11	0.33	1.00	0.40
bs	0.11	0.11	0.33	1.00	0.40
bc	0.11	0.11	0.33	1.00	0.40
bu	0.11	0.11	0.33	0.78	0.36
aq	0.11	0.11	0.33	0.78	0.36
av	0.11	0.11	0.33	0.78	0.36
dq	0.33	0.11	0.33	0.78	0.40
ai	0.11	0.11	0.33	1.00	0.40
bh	0.11	0.33	0.78	0.56	0.55
bi	0.33	0.33	0.78	0.56	0.60
cn	0.78	1.00	1.00	0.11	0.78
ck	0.56	0.56	0.78	0.33	0.62
an	1.00	0.56	1.00	0.33	0.82
ds	0.78	0.78	0.78	0.56	0.73
by	1.00	0.78	0.78	0.11	0.69
do	0.56	0.56	0.78	0.78	0.71
cx	0.56	0.56	0.33	0.78	0.49
cu	0.33	0.33	0.33	0.56	0.38
cv	0.33	0.33	0.33	0.56	0.38
cy	0.78	0.33	0.78	0.78	0.73
cb	0.78	0.78	1.00	0.56	0.84
ak	0.33	0.11	0.33	0.56	0.36
de	0.33	0.11	0.33	0.56	0.36
cj	1.00	1.00	0.33	0.11	0.49
ao	0.78	0.78	1.00	0.33	0.80
dj	0.33	0.78	1.00	0.33	0.71
al	0.11	0.11	0.33	1.00	0.40
am	0.11	0.11	0.33	1.00	0.40
bf	0.56	0.56	0.33	0.56	0.44
ar	0.11	0.33	0.78	1.00	0.64
as	0.11	0.33	0.33	0.56	0.33
bq	0.11	0.11	0.33	0.78	0.36
ch	0.11	0.11	0.33	1.00	0.40
af	0.33	0.33	0.33	0.56	0.38
ae	0.56	0.33	0.33	0.56	0.42
bg	0.33	0.11	0.33	0.33	0.31
br	0.56	0.56	0.33	0.33	0.40

Table I. Sample ABC Ratings by Criteria and Final Scores

Downloaded from http://www.emerald.com/ijopm/article-pdf/13/9/29/975952/01443579310043619.pdf by BI Norwegian Business School user on 26 February 2026
© Emerald Backfiles 2007

Part Code	Unit cost (0.20)	Lead time (0.10)	Downtime (0.50)	Demand (0.20)	Final score
ce	0.78	0.78	0.78	0.33	0.69
cf	0.78	0.78	0.78	0.33	0.69
cc	0.56	0.56	0.78	0.78	0.71
cg	1.00	1.00	1.00	0.56	0.91
cd	1.00	1.00	1.00	0.11	0.82
dd	0.33	0.11	0.33	1.00	0.44
au	0.33	0.11	0.33	1.00	0.44
dn	0.11	0.11	0.33	1.00	0.40
cm	0.11	0.33	0.33	0.78	0.38
ci	0.11	0.11	0.33	1.00	0.40
di	0.11	0.11	0.33	0.78	0.36
ah	0.11	0.33	0.33	0.56	0.33
dt	0.33	0.33	0.33	0.78	0.42
cl	0.11	0.33	0.33	0.56	0.33
dm	0.33	0.56	0.33	0.56	0.40
ct	0.11	0.56	0.33	0.56	0.36
aa	0.11	0.33	0.78	0.56	0.55
ac	0.56	0.78	0.78	0.78	0.73
cr	0.56	0.78	0.78	0.78	0.73
bd	0.33	0.11	0.33	0.56	0.36
be	0.33	0.33	1.00	0.56	0.71
cq	0.33	0.11	0.78	0.78	0.62
dl	0.78	0.33	0.33	0.56	0.47
dk	0.78	0.33	0.33	0.56	0.47
ad	0.56	0.11	0.78	1.00	0.71
aj	0.56	0.11	0.78	0.78	0.67
bk	0.56	0.33	0.78	1.00	0.73
bl	0.78	0.56	0.78	0.56	0.71
bm	0.78	0.56	0.78	0.56	0.71
cz	0.78	0.33	0.78	0.78	0.73
bo	0.56	0.33	0.78	0.33	0.60
da	0.11	0.11	0.33	0.56	0.31
db	0.11	0.33	0.33	0.78	0.38
aw	0.11	0.11	0.33	0.56	0.31
bx	0.11	0.11	0.33	1.00	0.40
bc	0.11	0.33	0.33	0.33	0.29
ap	0.11	0.11	0.78	1.00	0.62
ay	0.11	0.33	0.78	1.00	0.64
az	0.33	0.33	1.00	0.56	0.71
ba	0.33	0.78	1.00	0.78	0.80
dc	0.33	0.33	0.33	0.56	0.38
bp	0.33	0.11	0.33	1.00	0.44
cw	0.33	0.56	0.78	0.56	0.62
ca	0.56	0.11	0.33	0.78	0.44
dg	0.11	0.33	0.33	0.78	0.38
bn	0.11	0.33	1.00	0.56	0.67
dv	0.33	0.56	1.00	1.00	0.82
bt	0.11	0.33	0.33	0.33	0.29

Table I. (*Continued*)

Downloaded from http://www.emerald.com/ijopm/article-pdf/13/9/29/975952/01443579310043619.pdf by BI Norwegian Business School user on 26 February 2026
© Emerald Backfiles 2007

by six. The actual figures used were developed based on company records provided by management (see Appendix).

The simulation model is programmed in PASCAL and run on a Prime computer. The program uses discrete event logic and starts from a steady state. As demand for a part arrives, the corresponding part is released and the system updates itself. The simulation is event-oriented and advances the clock to the next event — demand for a part, order release, order receipts, or end of simulation. The model simulates a dynamic environment for a spare parts stockroom in which there is a continuous flow of parts demanded and delivered stochastically.

The spare part-ordering simulation follows the current inventory control policies. Therefore in the experiment we replenish our Class A SKUs as demand arises, maintaining a safety stock level of 0 or 1 units. The B items are ordered in batches of optimal order quantity at the reorder points, with no additional safety stock. Finally, C items are ordered four times a year, if the amount on-hand is less than the order quantity (the expected demand for three months), with no additional safety stock.

Each observation (replication) consisted of simulating five years (1,826 days) of operation. On average, 13,237 spare parts were demanded during the five years. This compares favourably with the actual demand for the same items in the last five years. On completion of five years, three performance criteria were collected:

(1) Downtime resulting from stock-out of critical item (days).

(2) Average inventory investment (dollars).

(3) Total ordering cost (dollars).

A total of 100 observations was collected for each factor-level combination. The average execution time per observation was approximately four CPU seconds.

Results

The most important evaluative criterion for management was the cost versus the benefits of the proposed classification system. Comparison of the inventory-associated costs, ordering and holding, requires the specification of an inventory holding rate. The firm assumes a rate of 20 per cent of the item value per year. Under this assumption, the average inventory-associated costs per year using the AHP model were $14,231.51 and $14,059.28 under dollar usage. The difference is $172.22 per year in favour of the dollar usage method. However, this saving is minimal compared with the savings in downtime using the AHP model. The proposed AHP model resulted in an average of 12.65 days of downtime compared with 273.10 days with dollar usage, over the five-year simulation. This reduction of approximately 50 days per year will overshadow the modest increase in inventory costs.

Although the firm assumed a holding rate of 20 per cent, other holding rates were also considered, and the results are shown in Table II. As seen, even at the least favourable rate, 10 per cent, the difference in inventory costs is only $876.33 per year. The break-even holding rate is approximately 22.4 per cent, so that a rate higher than this results in favouring AHP in terms of inventory cost, while maintaining the benefit of reduced downtime.

Downloaded from http://www.emerald.com/ijopm/article-pdf/13/9/29/975952/01443579310043619.pdf by BI Norwegian Business School user on 26 February 2026
© Emerald Backfiles 2007

Analysis was conducted to see if the difference in performance results was statistically significant. The experiment uses a two-factor experiment design in comparing the performance of the proposed inventory classification model with the traditional dollar usage classification, as the safety stock level for class A items changes. One factor is the method used, the proposed AHP model versus the dollar usage model. The second factor is safety stock (SS) for A items. This factor has two levels (SS = 0,1). The data are the average and standard deviation of 100 replications (see Table III).

Since changing the level of the safety stock from one to two for A items does not affect the mean downtime or ordering cost using either model, but does increase the average inventory investment, we conclude that increasing safety stock beyond one unit is a waste of capital resources. This means that analysis of the classification type and safety stock level is based on the three performance measures mentioned in the previous section, and only two levels of the safety stocks. Table IV shows the analysis of variance (ANOVA) of three performance measures for our AHP model and the traditional dollar usage model at two levels of safety stock. The ANOVA model assumes that the data within a treatment mechanism are independent and normally distributed, and that

Holding rate	AHP ($)	Dollar usage	Difference ($)
0.10	11,102.75	10,226.42	876.33
0.14	12,354.25	11,759.57	594.68
0.16	12,980.00	12,526.14	453.86
0.18	13,605.75	13,292.71	313.04
0.20	14,231.51	14,059.28	172.23
0.22	14,857.26	14,825.86	31.40
0.24	15,483.01	15,592.43	-109.41
0.26	16,108.76	16,359.10	-250.23
0.28	16,734.51	17,125.57	-391.06
0.30	17,360.27	17,892.14	-531.87

Table II. Comparison of Inventory Costs at Different Holding Rates for AHP versus Dollar-usage Method

Safety stock (SS)		Downtime (day)		Ordering cost ($)		Averge inventory investment ($)	
		AHP	Dollar usage	AHP	Dollar usage	AHP	Dollar usage
0	Mean	12.65	273.10	39,869.92	31,967.80	31,287.61	38,328.79
	Standard deviation	5.77	43.17	420.17	982.05	563.21	505.79
1	Mean	0.00	265.99	39,842.53	31,908.86	51,420.81	59,357.96
	Standard deviation	0.00	39.66	362.17	993.28	520.05	489.97
2	Mean	0.00	265.99	39,842.53	31,908.86	71,886.86	80,276.65
	Standard deviation	0.00	39.66	362.17	993.28	519.44	490.74

Table III. Performance Statistics for AHP versus Dollar-usage Classifications

Downloaded from http://www.emerald.com/ijopm/article-pdf/13/9/29/975952/01443579310043619.pdf by BI Norwegian Business School user on 26 February 2026
© Emerald Backfiles 2007

Table IV. ANOVA Results for Downtime, Ordering Cost, and Average Inventory Investment for AHP and Dollar Usage Methods of Inventory Classifications

	Source			
	Classification method	Safety stock	Classification method	Safety stock
Degree of freedom	1	1	1	
Downtime				
Anova SS	6,928,426.83	9,760.35	769.65	
F value	7,987.17	11.25	0.89	
Pr>F	0.0001	0.0009	0.3468	
Ordering cost				
Anova SS	6,269,307,859.21	186,326.62	24,873.60	
F value	11,102.32	0.33	0.04	
Pr>F	0.0001	0.5660	0.8339	
Average inventory investment				
Anova SS	5,608,765,031.73	42,358,533,207.67	20,069,346.23	
F value	20,704.93	99,999.99	74.09	
Pr>F	0.001	0.0001	0.0001	

data across the population have homogeneous variance. Naylor *et al.* [21] indicate that the independence of the pseudo-random number used in simulation satisfies the independence assumption. Furthermore, ANOVA is robust; that is, it is quite insensitive to departure from assumptions of normality and homoscedasticity.

Table IV indicates, as expected, that changing the level of the safety stock from zero to one significantly affects the mean downtime ($p<0.0009$) and average inventory investment ($p<0.0001$). However, changes in ordering cost ($p<0.5660$) are not significant.

Furthermore, it can be seen from Table IV that the performance of the proposed AHP-based model with respect to downtime resulting from stock-out is significantly better ($p<0.0001$) than the dollar usage classification for the two levels of safety stocks. This output measure is the single most important performance criterion for continuity and efficiency of manufacturing operations. The costs of downtime are generally acknowledged to be much greater than those related to inventory. Downtime should be avoided whenever possible.

In addition, the difference in average inventory investment is statistically significant ($p<0.0001$) and lower in our proposed AHP model, while ordering cost is higher ($p<0.0001$). The dollar usage classification saves around $8,000 on ordering costs over five years, but the AHP model yields far superior savings in factory uptime, plus reduction of average inventory investment.

There is no statistical interaction between ABC classification methods and level of safety stocks in any of the performance measures, except the average inventory investment ($p<0.0001$) (see Table IV). Figure 3 presents a profile plot of average inventory investment for two safety stock levels under the

Downloaded from http://www.emerald.com/ijopm/article-pdf/13/9/29/975952/01443579310043619.pdf by BI Norwegian Business School user on 26 February 2026
© Emerald Backfiles 2007

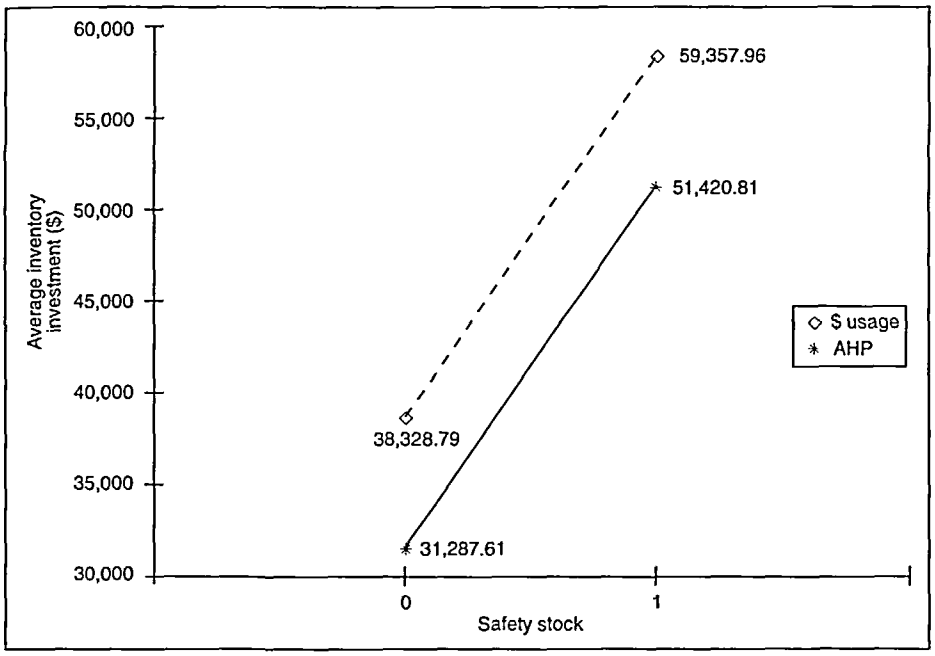

Figure 3. Average Inventory Investment for Different Levels of Safety Stock

alternative classification methods. The plot shows that the two lines are not far from parallel, which indicates that the interaction is of no practical importance.

The management of the pharmaceutical company has not yet implemented the AHP classification system at this time. Nevertheless, they agreed that the proposed model could have a significant impact on the productivity of their operations.

Summary and Conclusions

Besides dollar usage there are other criteria which are important in managing SKUs. This article proposes a new multicriteria model for classifying inventory items. The proposed methodology, which is based on AHP, can incorporate all relevant qualitative, as well as quantitative, criteria for classifying SKUs. An example is provided by using real data from the maintenance department stockroom of a pharmaceutical company to demonstrate an application of the model. Through a series of simulation experiments, the article shows how the proposed model can improve the inventory control in this company. The results indicate that the use of the proposed AHP-based classification method increases the ordering cost marginally, but substantially reduces downtime and average inventory investment.

This study raises several interesting issues which may warrant further research. Though this article has demonstrated the use and the benefits of the proposed model through a real-life example, evaluation of the proposed model using a detailed field study is desirable. In addition, comparison of the proposed model with other multicriteria methods seems appropriate. Finally, this approach might be extended to other multicriteria classifications in manufacturing.

Downloaded from http://www.emerald.com/ijopm/article-pdf/13/9/29/975952/01443579310043619.pdf by BI Norwegian Business School user on 26 February 2026
© Emerald Backfiles 2007

References

1. Silver, E.A. and Peterson, R., *Decision Systems for Inventory Management and Production Planning*, 2nd ed., Wiley, New York, NY, 1985.
2. Saaty, T.L., *The Analytic Hierarchy Process*, McGraw-Hill, New York, NY, 1980.
3. Campbell, K.L., "Inventory Turns and ABC-Analysis — Outmoded Textbook Concepts?", *APICS Eighteenth Annual International Conference Proceedings*, San Diego, CA, 1975, pp. 421-32.
4. Zimmerman, G.W., "The ABCs of Vilfredo Pareto", *Production and Inventory Management*, Vol. 16 No. 3, 1975, pp. 1-9.
5. Flores, B.E. and Whybark, D.C., "Implementing Multiple Criteria ABC Analysis", *Journal of Operations Management*, Vol. 7 Nos. 1 & 2, 1987, pp. 79-84.
6. Flores, B.E. and Whybark, D.C., "Multiple Criteria ABC Analysis", *International Journal of Operations & Production Management*, Vol. 6 No. 3, 1986, pp. 38-45.
7. Ernst, R. and Cohen, M.A., "Operations-related Groups (ORGs): A Clustering Procedure for Production/Inventory Systems", *Journal of Operations Management*, Vol. 9 No. 4, 1990, pp. 574-98.
8. Cohen, M.A. and Ernst, R., "Multi-item Classification and Generic Inventory Stock Control Policies", *Production and Inventory Management*, Vol. 29 No. 3, 1988, pp. 6-8.
9. Zahedi, F., "The Analytic Hierarchy Process — A Survey of the Method and Its Application", *Interfaces*, Vol. 16 No. 4, 1986, pp. 96-108.
10. Partovi, F.Y., "An Analytic Hierarchy Approach to Activity-based Costing", *International Journal of Production Economics*, Vol. 22 No. 2, 1991, pp. 151-61.
11. Kleindorfer, P.R. and Partovi, F.Y., "Integrating Manufacturing Strategy and Technology Choice", *European Journal of Operations Research*, Vol. 47 No. 2, 1990, pp. 214-24.
12. Madu, C.N. and Georgantzas, N.C., "Strategic Thrust of Manufacturing Automation Decision: A Conceptual Framework", *IIE Transactions*, Vol. 23 No. 2, 1991, pp. 138-48.
13. Cambron, K.E. and Evans, G.W., "Layout Design Using the Analytic Hierarchy Process", *Computers & Industrial Engineering*, Vol. 20 No. 2, 1991, pp. 211-29.
14. Partovi, F.Y., Burton, J. and Banerjee, A., "Application of Analytical Hierarchy Process in Operation Management", *International Journal of Operations & Production Management*, Vol. 10 No. 3, 1990, pp. 5-19.
15. Saaty, T.L., "How to Make a Decision: The Analytic Hierarchy Process", *European Journal of Operations Research*, Vol. 48 No. 1, 1990, pp. 9-26.
16. Saaty, T.L., France, J.W. and Valentine, K.R., "Modeling the Graduate Business School Admissions Process", *Socio-Economic Planning Sciences*, Vol. 25 No. 2, 1991, pp. 155-62.
17. Stewart, W.R. Jr and Horowitz, E.R., "Environmental Factor Weighting at the Federal Energy Regulatory Commission", *Socio-Economic Planning Sciences*, Vol. 25 No. 2, 1991, pp. 123-32.
18. Forman, E.H. and Saaty, T.L., *Expert Choice Software Package for IBM PC*, Expert Choice, Pittsburgh, PA, 1991.
19. Saaty, T.L., "An Explanation of the AHP in Reply to the Paper, 'Remarks on the Analytic Hierarchy Process' ", *Management Science*, Vol. 36 No. 3, 1990, pp. 259-68.
20. Saaty, T.L., "Ratio Scales Derived from Perturbations of Consistent Judgments", *Behaviourmetrika*, No. 28, 1990, pp. 1-21.
21. Naylor, T.H., Balintfy, J.L., Burdick, D.S. and Kong, C., *Computer Simulation Techniques*, John Wiley & Sons, New York, NY, 1968.

Downloaded from http://www.emerald.com/ijopm/article-pdf/13/9/29/975952/01443579310043619.pdf by BI Norwegian Business School user on 26 February 2026
© Emerald Backfiles 2007

Appendix. Listing of Spare Part SKUs for a Maintenance Department of a Pharmaceutical Company

Item code	Unit cost ($/unit)	Procurement ($/lot)	Demand (units/year)	Lead time (days)
aa	1.50	0.07	11 - 20	3 - 5
ab	27.00	1.00	41 - 75	15 - 35
ac	56.00	2.00	21 - 40	15 - 35
ad	54.00	2.50	41 - 75	1 - 2
ae	103.00	5.00	11 - 20	3 - 5
af	15.00	0.69	11 - 20	3 - 5
ag	0.10	0.05	41 - 75	1 - 2
ah	0.10	0.05	11 - 20	3 - 5
ai	0.10	0.50	41 - 75	1 - 2
aj	34.00	2.00	21 - 40	1 - 2
ak	48.00	2.00	11 - 20	1 - 2
al	0.10	0.05	41 - 75	1 - 2
am	0.10	0.05	41 - 75	1 - 2
an	600.00	25.00	3 - 10	6 - 14
ao	400.00	20.00	3 - 10	15 - 35
ap	0.10	0.05	41 - 75	1 - 2
aq	0.10	0.05	21 - 40	1 - 2
ar	3.75	0.20	41 - 75	3 - 5
as	0.10	0.05	11 - 20	3 - 5
at	100.00	5.00	21 - 40	15 - 35
au	13.00	0.40	41 - 75	1 - 2
av	0.10	0.05	21 - 40	1 - 2
aw	0.10	0.05	11 - 20	1 - 2
ax	150.00	7.00	11 - 20	3 - 5
ay	7.00	0.30	41 - 75	3 - 5
az	10.00	0.50	11 - 20	3 - 5
ba	30.00	1.50	21 - 40	15 - 35
bb	4,570.00	150.00	3 - 10	36 - 70
bc	0.10	0.05	41 - 75	1 - 2
bd	4.50	0.25	11 - 20	1 - 2
be	5.00	0.25	11 - 20	3 - 5
bf	70.00	3.00	11 - 20	6 - 14
bg	18.00	0.75	3 - 10	1 - 2
bh	3.75	0.15	11 - 20	3 - 5
bi	5.00	0.01	11 - 20	3 - 5
bj	0.10	0.05	41 - 75	1 - 2
bk	75.00	3.00	41 - 75	3 - 5
bl	175.00	8.00	11 - 20	6 - 14
bm	300.00	15.00	11 - 20	6 - 14
bn	1.00	0.05	11 - 20	3 - 5
bo	87.00	4.00	3 - 10	3 - 5
bp	26.00	1.00	41 - 75	1 - 2
bq	0.10	0.05	21 - 40	1 - 2
br	120.00	5.00	3 - 10	6 - 14
bs	0.10	0.05	41 - 75	1 - 2
bt	0.10	0.05	3 - 10	3 - 5
bu	0.10	0.05	21 - 40	1 - 2

Table AI.

Downloaded from http://www.emerald.com/ijopm/article-pdf/13/9/29/975952/01443579310043619.pdf by BI Norwegian Business School user on 26 February 2026
© Emerald Backfiles 2007

Item code	Unit cost ($/unit)	Procurement ($/lot)	Demand Units/year	Lead time (days)
bv	0.10	0.05	41 - 75	1 - 2
bx	0.10	0.50	41 - 75	1 - 2
by	730.00	17.00	1 - 2	15 - 35
bz	0.10	0.50	41 - 75	1 - 2
ca	20.00	1.00	21 - 40	3 - 5
cb	520.00	10.00	11 - 20	15 - 35
cc	161.00	8.00	21 - 40	6 - 14
cd	7,132.00	300.00	1 - 2	36 - 70
ce	400.00	20.00	3 - 10	15 - 35
cf	249.00	10.00	3 - 10	15 - 35
cg	2,614.00	100.00	11 - 20	36 - 70
ch	0.10	0.05	41 - 75	1 - 2
ci	0.10	0.05	41 - 75	1 - 2
cj	1,425.00	50.00	1 - 2	36 - 70
ck	72.00	3.00	3 - 10	6 - 14
cl	0.10	0.05	11 - 20	3 - 5
cm	0.10	0.05	21 - 40	3 - 5
cn	105.00	5.00	1 - 2	36 - 70
cq	12.00	0.50	21 - 40	1 - 2
cr	108.00	5.00	21 - 40	15 - 35
cs	0.10	0.05	21 - 40	1 - 2
ct	3.40	0.15	11 - 20	6 - 14
cu	40.00	2.00	11 - 20	3 - 5
cv	40.00	2.00	11 - 20	3 - 5
cw	15.00	0.75	11 - 20	6 - 14
cx	65.00	3.00	21 - 40	6 - 14
cy	116.00	30.00	21 - 40	3 - 5
cz	129.00	6.00	21 - 40	3 - 5
da	0.10	0.05	11 - 20	1 - 2
db	1.00	0.05	21 - 40	3 - 5
dc	12.00	0.60	11 - 20	3 - 5
dd	8.50	0.40	41 - 75	1 - 2
de	10.00	0.50	11 - 20	1 - 2
df	2,995.00	100.00	1 - 2	36 - 70
dg	0.10	0.05	21 - 40	3 - 5
di	0.10	0.05	21 - 40	1 - 2
dj	35.00	2.00	3 - 10	15 - 35
dk	363.00	15.00	11 - 20	3 - 5
dl	150.00	7.00	11 - 20	3 - 5
dm	19.00	0.07	11 - 20	6 - 14
dn	0.10	0.05	41 - 75	1 - 2
do	75.00	3.00	21 - 40	6 - 14
dp	30.00	7.00	21 - 40	3 - 5
dq	34.00	1.50	21 - 40	1 - 2
dr	330.00	15.00	11 - 20	6 - 14
ds	140.00	7.00	11 - 20	15 - 35
dt	5.80	0.25	21 - 40	3 - 5
du	882.00	40.00	3 - 10	15 - 35
dv	17.00	0.50	41 - 75	6 - 14

Table AI.
(*Continued*)

Downloaded from http://www.emerald.com/ijopm/article-pdf/13/9/29/975952/01443579310043619.pdf by BI Norwegian Business School user on 26 February 2026
© Emerald Backfiles 2007